IMPACT OF RABIES IN DEVELOPING COUNTRIES

MATH 574:

MATHEMATICAL MODELLING OF INFECTIOUS DISEASES

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Categories:

1. Project Topic Decision
2. Model Development
3. Creation of Research Objectives
4. Proposal Writing/Editing
5. Primary Research
6. Supplemental Research
7. Data Research/Collection
8. Model Analysis
9. Model Coding/Editing
10. Parameter Estimation
11. Sensitivity Analysis
12. Presentation Creation/Editing
13. Report Writing/Editing

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1. INTRODUCTION

1.1: BACKGROUND

Rabies is a viral disease that exists in many mammal species, including humans. The first documented case of rabies dates back over 4,000 years [1], making it one of the world's oldest diseases [2]. The disease primarily occurs in wild animals such as dogs, bats, foxes, and skunks, and is most often spread when an infected animal bites or scratches a susceptible host. Over 99% of human infections can be traced back to bites from domesticated dogs [3]. Symptoms of rabies generally occur between 20 days and 3 months and include fever, headaches, fatigue, restlessness, anxiety, hallucinations, hydrophobia, difficulty swallowing and paralysis [6]. Once infected, rabies is nearly 100% fatal to humans; however, it is also almost 100% preventable through prompt treatment or vaccination [3]. Deaths can be prevented with prompt post-exposure prophylaxis (PEP), which restricts the virus from reaching the central nervous system [4]. However, despite the existence of these life-saving methods, rabies claims approximately 59,000 lives per year, primarily in Africa and Asia [3].

1.2: MOTIVATION

This project aims to model the transmission of rabies from infected dogs to humans in developing countries. Our model intends to identify the most important factors involved in transmission of rabies from stray dogs to humans. The overarching goal of this project is to find the best approach to minimizing the number of human deaths resulting from rabies. Our analysis will be used to suggest steps that individuals and/or government organizations can take to help reduce the spread of rabies and save human lives.

2. Model

2.1: Diagram

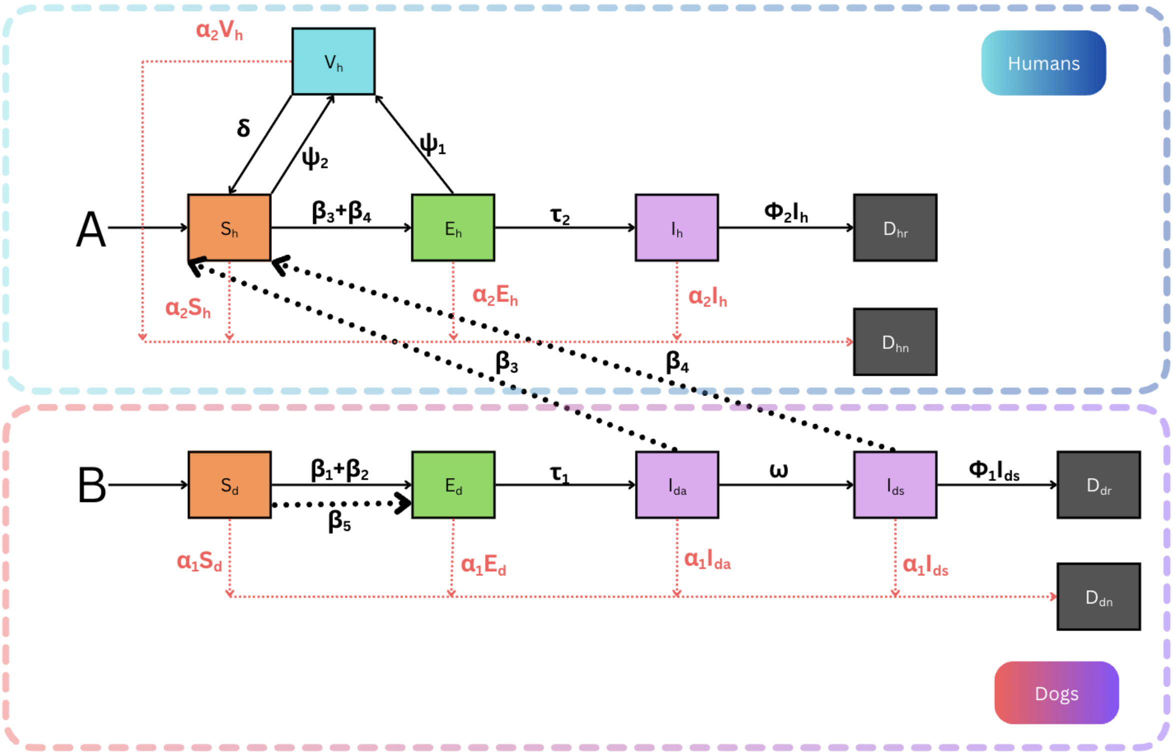

Figure 1: Model Diagram

2.2: Compartments And Notations

Notation	Description of Notation
B	Birth rate of stray dogs in India → S_h
A	Cumulative addition to the "susceptible" human population, including births → S_d
β_1	Transmission coefficient from I_{da} → S_d
β_2	Transmission coefficient from I_{ds} → S_d
β_3	Transmission coefficient from I_{da} → S_h
β_4	Transmission coefficient from I_{ds} → S_h
β_5	Background/natural transmission coefficient from other animals to dogs
τ_1	Incubation period from E_d → I_{da}
τ_2	Incubation period from E_h → I_h
ω	Progression rate from I_{da}→ I_{ds}
ψ_1	Rate at which exposed dogs are vaccinated (reactive vaccination) E_d → V_d
ψ_2	Vaccination rate for susceptible dogs S_d → V_d
ψ_3	Rate at which exposed humans are vaccinated E_h→ V_h
ψ_4	Rate at which susceptible humans are vaccinated S_h→ V_h
δ_1	Rate at which vaccinated dogs lose immunity V_d → S_d
δ_2	Rate at which vaccinated humans lose immunity V_h → S_h
α_1	Natural death rate for dogs → D_{dn}
α_2	Natural death rate for humans → D_{hn}
ϕ_1	Rabies-induced death rate for symptomatic dogs I_{ds} → D_{dr}
ϕ_2	Rabies-induced death rate for infected humans I_h → D_{hr}

Table 2: Model Notations

Compartment	Description of Compartment
S_d	"Susceptible" population of dogs
E_d	"Exposed" population of dogs
I_{da}	"Asymptomatically Infected" population of dogs
I_{ds}	"Symptomatically Infected" population of dogs
D_{dr}	"Deceased" population of dogs due to rabies
D_{dn}	"Deceased" population of dogs due to natural causes
S_h	"Susceptible" population of humans
V_h	"Vaccinated" population of humans
E_h	"Exposed" population of humans
I_h	"Infected" population of humans
D_{hr}	"Deceased" population of humans due to rabies
D_{hn}	"Deceased" population of humans due to natural causes

Table 3: Model Compartments

3. METHODOLOGY

3.1: PARAMETERS

Parameter	Description	Estimate	Reference
A	*Human birth rate (India)*	*16.55 per 1000 people i.e: 6.8014e+4*	https://www.macrotrends.net/global-metrics/countries/ind/india/birth-rate
B	*Dog birth rate (India)*	*3.9/100 dogs i.e: 3.994e+3*	https://www.hindustantimes.com/cities/lucknow-news/six-years-on-lucknow-halves-stray-dog-birth-rate-101755540834906.html
β_1*	Asymptomatic dogs to dogs infection rate	5.5185e-10	https://pmc.ncbi.nlm.nih.gov/articles/PMC7613728/

β_2*	Symptomatic dogs to dogs infection rate	8.4061e-10	https://pmc.ncbi.nlm.nih.gov/articles/PMC7613728/
β_3*	Asymptomatic to human infection rate	8.2878e-10	https://www.cmaj.ca/content/178/5/564 https://www.sciencedirect.com/science/article/pii/S1201971206000117
β_4*	Symptomatic to human infection rate	3.3016e-08	https://www.cmaj.ca/content/178/5/564 https://www.sciencedirect.com/science/article/pii/S1201971206000117
β_5*	Other rabid animals to dogs infection rate	8.0247e-08	https://pmc.ncbi.nlm.nih.gov/articles/PMC2728111/
τ_1	Dog incubation period	4 weeks on average (2-8 range)	https://www.smalldoorvet.com/learning-center/medical/rabies-in-dogs#incubation
τ_2	Human incubation period	2–3 months	https://www.who.int/news-room/fact-sheets/detail/rabies
ω	Symptomatic onset for dogs	3 days	https://www.petmd.com/dog/conditions/neurological/c_multi_rabies
ψ_1	Rate at which exposed humans are vaccinated (india)	0.02	https://www.cmaj.ca/content/178/5/564
ψ_2	Rate at which susceptible humans are vaccinated	1.3699e-05	[Estimated Value]
δ	Human vaccine immunity loss	2 years	https://www.nhs.uk/conditions/rabies/vaccination/
α_1	Indian street dog lifespan	13 years	https://furloved.co/blogs/news/indian-street-dogs-breed?srsltid=AfmBOor9gCg7zJz38aR0g1OrqBIYt8mDd1PHXvxeD4o9lIQR8OtEsGyL
α_2	Human life expectancy in india	3.86e-05	https://www.macrotrends.net/global-metrics/countries/ind/india/life-expectancy
ϕ_1	Time until death for dogs	10 days i.e: 0.1	https://www.petmd.com/dog/conditions/neurological/c_multi_rabies
ϕ_2	Time to death for people	7-14 days i.e.: 0.1429	https://www.canada.ca/en/public-health/services/diseases/rabies.html

N0_h	Population of india	1,463,865,525	https://www.worldometers.info/world-population/india-population/
S0_h	Initially susceptible people	1.5e9*0.99	Assumption that 1% of the population was already vaccinated due to pre-existing cases and pre-exposed vaccine
V0_h	Initially vaccinated people	1.5e9*0.01	Assumption that 1% of the population was already vaccinated due to pre-existing cases and pre-exposed vaccine
E0_h	Initially exposed people	0	Estimate for the purpose of demonstration
I0_h	Initially infected people	0	Estimate for the purpose of demonstration
N0_d	Population of stray dogs in india	37,37,9600	https://www.cmaj.ca/content/178/5/564
S0_d	Susceptible dogs	37,379,597	[Total Population - Initially asymptomatic infected dogs - Initially symptomatic infected dogs]
E0_d	Initially exposed dogs	0	[Estimated Values]
I0_da	Initially asymptomatic infected dogs	1	Estimated value to test the endemic nature of the disease
I0_ds	Initially symptomatic infected dogs	2	Estimated value to test the endemic nature of the disease

Table 4: Parameter Estimation

Note: All the parameters with '*' were not found because of limited data so they were fitted using Bayesian Inference.

3.2: EQUATIONS

3.2.1 EQUATIONS FOR DOGS

$$\frac{dS_d}{dt} = B - \beta_1 S_d I_{da} - \beta_2 S_d I_{ds} - \beta_5 S_d - \alpha_1 S_d \tag{1}$$

$$\frac{dE_d}{dt} = \beta_1 S_d I_{da} + \beta_2 S_d I_{ds} + \beta_5 S_d - (\tau_1 + \alpha_1) E_d \tag{2}$$

$$\frac{dI_{da}}{dt} = \tau_1 E_d - (\omega + \alpha_1) I_{da} \tag{3}$$

$$\frac{dI_{ds}}{dt} = \omega I_{da} - (\phi_1 + \alpha_1) I_{ds} \tag{4}$$

$$\frac{dD_{dr}}{dt} = \phi_1 I_{ds} \tag{5}$$

3.2.2 EQUATIONS FOR HUMANS

$$\frac{dS_h}{dt} = A + \delta V_h - \beta_3 S_h I_{da} - \beta_4 S_h I_{ds} - (\psi_2 + \alpha_2) S_h \tag{6}$$

$$\frac{dV_h}{dt} = \psi_1 E_h + \psi_2 S_h - (\delta + \alpha_2) V_h \tag{7}$$

$$\frac{dE_h}{dt} = \beta_3 S_h I_{da} + \beta_4 S_h I_{ds} - (\psi_1 + \tau_2 + \alpha_2) E_h \tag{8}$$

$$\frac{dI_h}{dt} = \tau_2 E_h - (\phi_2 + \alpha_2) I_h \tag{9}$$

$$\frac{dD_{hr}}{dt} = \phi_2 I_h \tag{10}$$

3.3: Model Assumptions

The model is based on the following assumptions:

1. All transmission rates are constant.
2. No human-to-human transmission.
3. No human-to-dog transmission.
4. Humans can only get rabies from stray dogs.
5. Immunity loss is constant for all vaccinated humans.
6. Rabies vaccines and treatment are 100% successful.
7. Stray dogs do not get vaccinated.
8. The progression of the disease is constant.
9. Rabies mortality rate is 100% for dogs and humans.

3.4: Data Collection

Most of our parameters were extracted from the literature surrounding the rabies disease. The data we collected was primarily based in India to keep our calculations specific and aid in the accuracy of our parameter fitting.

3.5: Parameter Fitting

Rabies transmission rates (β) were particularly challenging to find reliable numbers for, so we used the Bayesian parameter fitting approach to estimate these 5 rates. Due to data limitations, we chose to fit our parameters on a single number: the estimated average number of rabies-related deaths in India per year. It is also important to note that due to time constraints, we were forced to prioritize speed over accuracy in our fitting algorithm.

3.6: Sensitivity Analysis

Sensitivity analysis helps determine the parameters that most strongly drive the model outcomes. In this case, we are interested in studying how our parameters impact the number of human deaths caused by rabies. To investigate this, we conducted a time-dependent sensitivity analysis on the following variables of interest:

1. Birth rate of stray dogs (B)
2. Rate of Asymptomatic transmission (β_1)
3. Rate of Symptomatic transmission (β_2)
4. Vaccination rate of Susceptible humans(ψ_2)
5. Vaccination/treatment rate of Exposed humans(ψ_1)

This analysis identifies the magnitude and direction of the link between these parameters and the resulting number of human deaths over 20 years. With a deeper understanding of the primary forces within our model, we can make more accurate suggestions on how to reduce the impact of rabies on human lives.

4. RESULTS

4.1: PARAMETER FITTING

To compute the transmission rates of rabies amongst our populations, we relied on Bayesian parameter fitting methods. We used the estimated average number of rabies-related deaths in India (25,222) per year to fit our model. A quick look at figures a-e makes it apparent that we faced the common problem of non-identifiability. This means any of the 'spikes' in these figures represent a parameter value that will produce a good fit. Unfortunately, we were unable to find a solution to this problem. Therefore, we used the value corresponding to the most likely parameter (i.e. the highest point in each figure) as our transmission rates.
The best estimate for Median values of transmission coefficient (β_s) were found to be:

β_1 = 1.9914e-10; β_3 = 4.2361e-08; β_5 = 2.7289e-08;
β_2 = 7.3141e-10; β_4 = 2.6464e-08;

The best estimate for MAP values of transmission coefficient (β_s) were found to be:

β_1 = 5.5185e-10; **β_3 = 8.2878e-10;** **β_5 = 8.0247e-08;**
β_2 = 8.4061e-10; **β_4 = 3.3016e-08;**

As MAP values gave us a better estimate of the transmission rate as it provides the single most probable parameter value. Our model is density dependent so MAP provides a better value estimation as it is easy to compute if you can evaluate the posterior density.

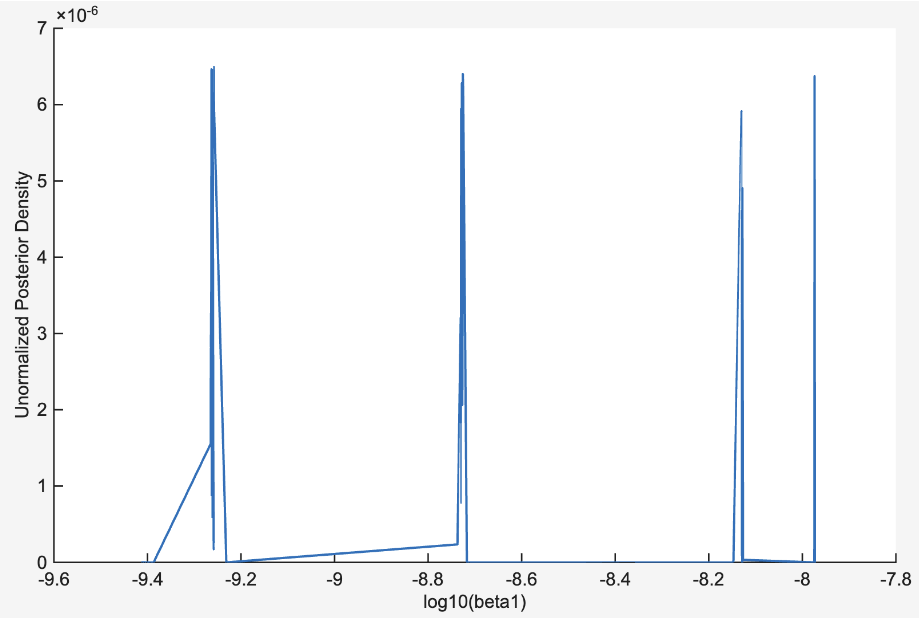

Figure 2: log10(β_1), $\beta_1 = 5.5185\ e^{-10}$

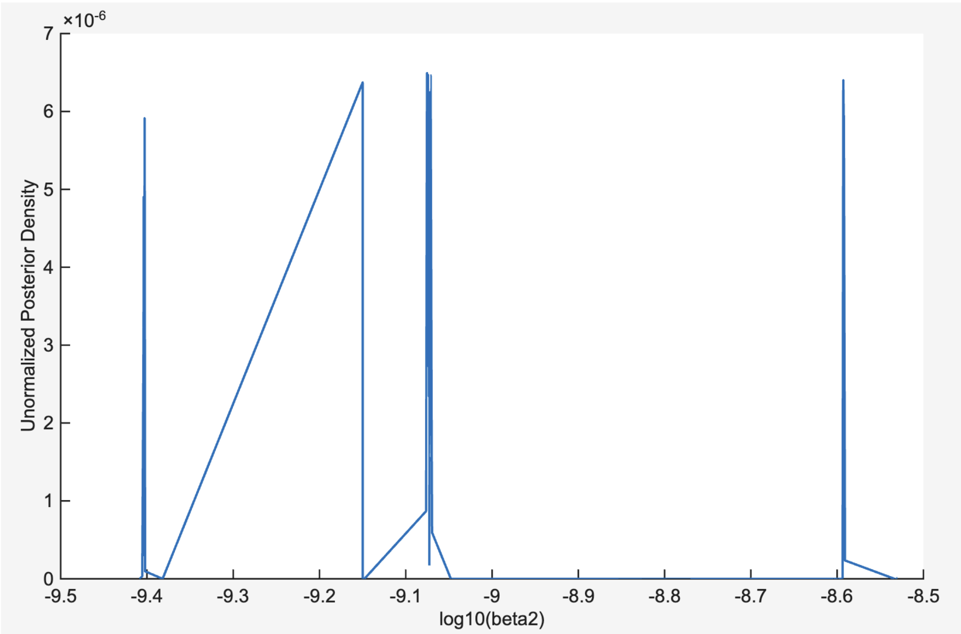

Figure 3: log10(β_2), $\beta_2 = 8.4061\ e^{-10}$

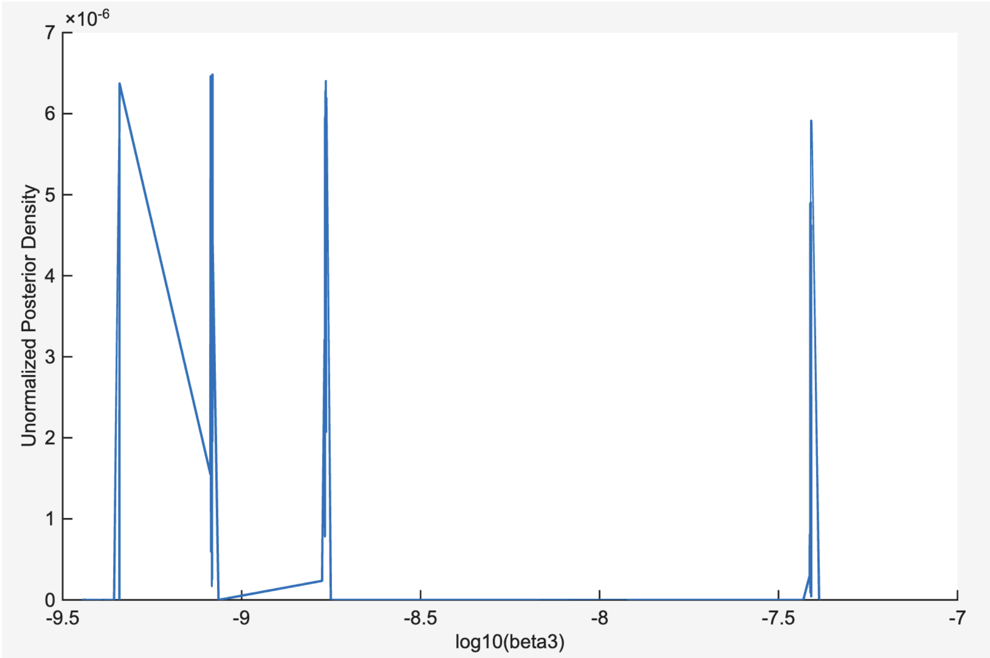

Figure 4: $\log10(\beta_3)$, $\beta_3 = 8.2878\ e^{-10}$

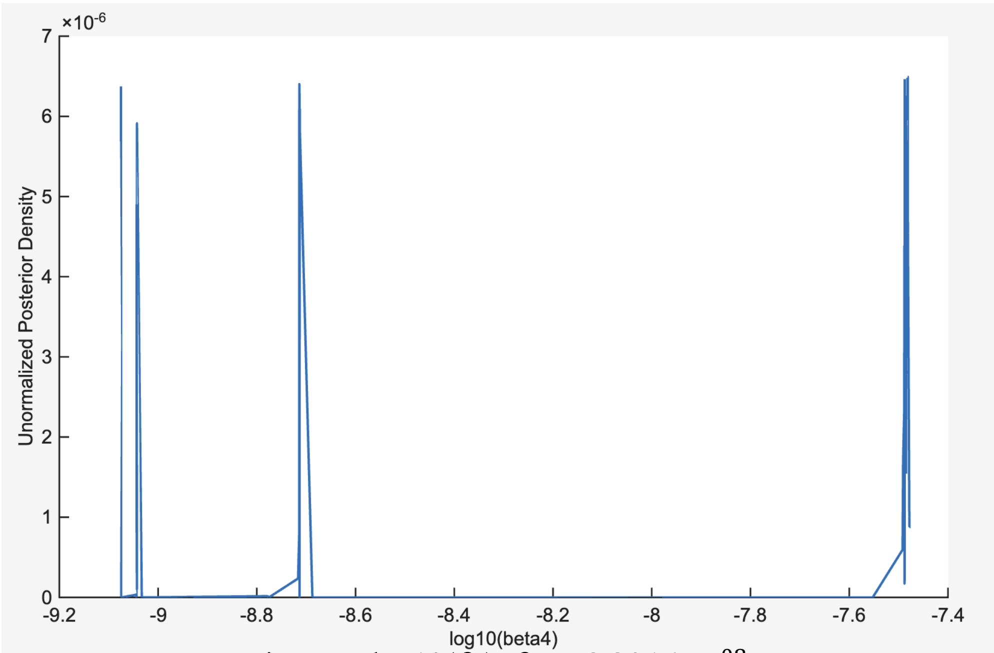

Figure 5: $\log10(\beta_4)$, $\beta_4 = 3.3016\ e^{-08}$

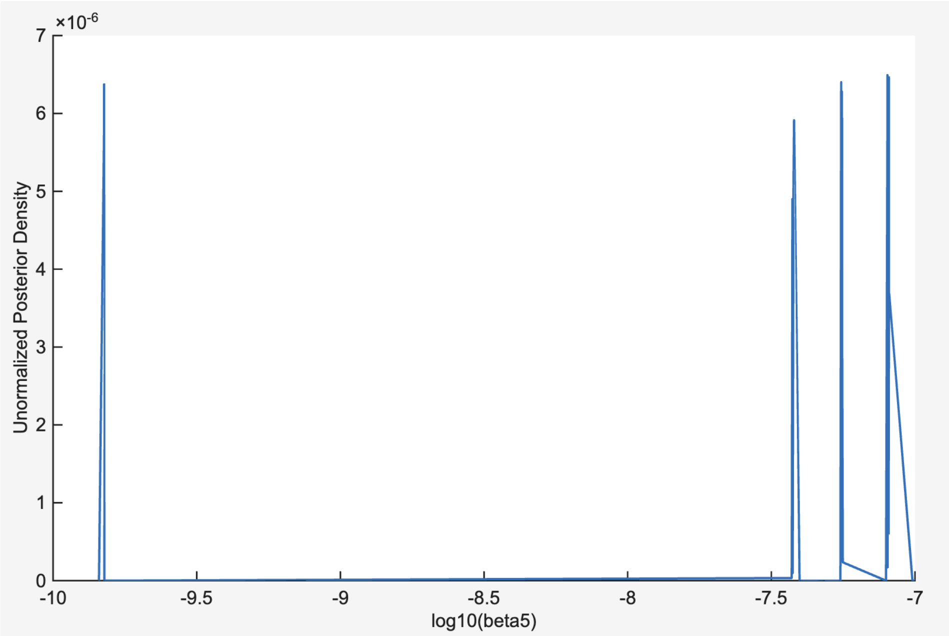

Figure 6: $\log10(\beta_5)$, $\beta_5 = 8.0247\ e^{-08}$

Figure below displays a plot of our model and the expected number of deaths we used to estimate our parameters. The blue model line passes directly through the black circle, which indicates a good fit.

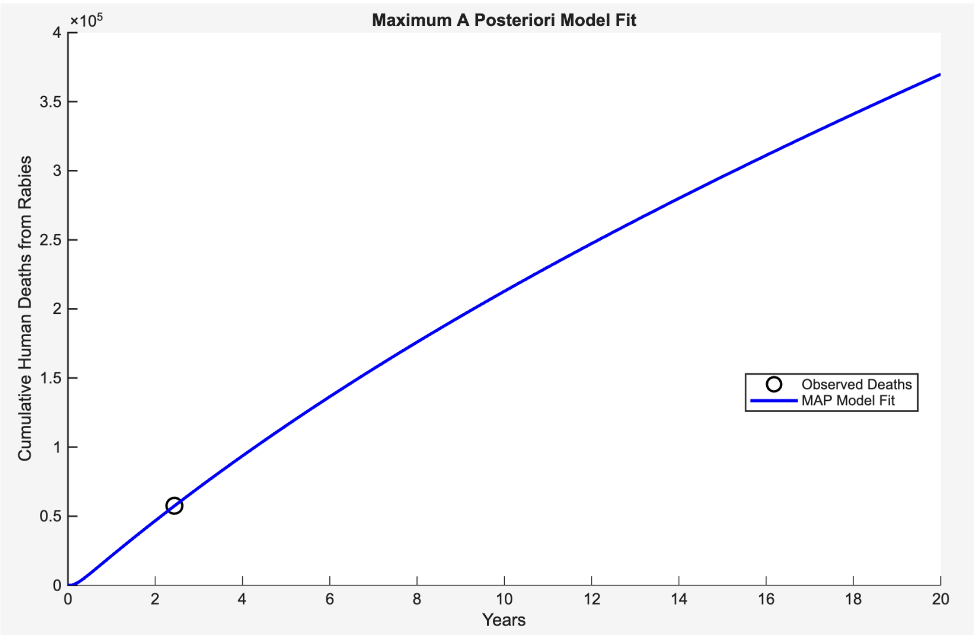

Figure 7: Bayesian Model Fit

4.2: Model Outputs

We plotted separate graphs of the population of dogs and humans respectively over a 2 year period (figures 8, 9, 10, 11). These plots highlight the overall landscape of rabies for each population in India. It is clear from both figures that rabies does not have a significant impact on the overall population of its hosts over a short period.

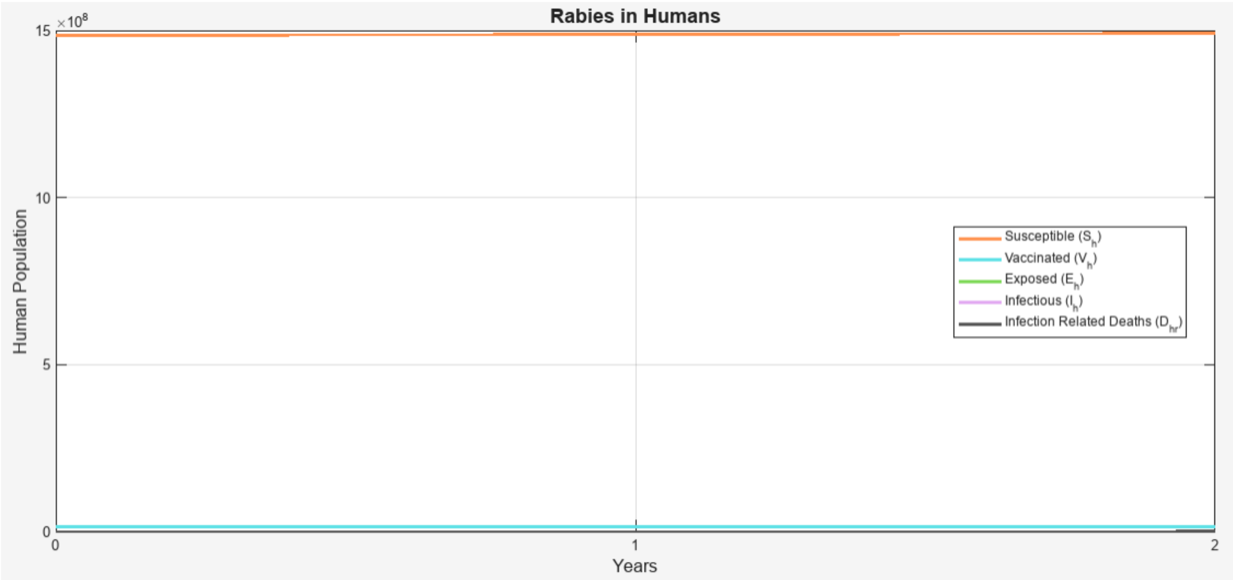

Figure 8: SIR for Humans

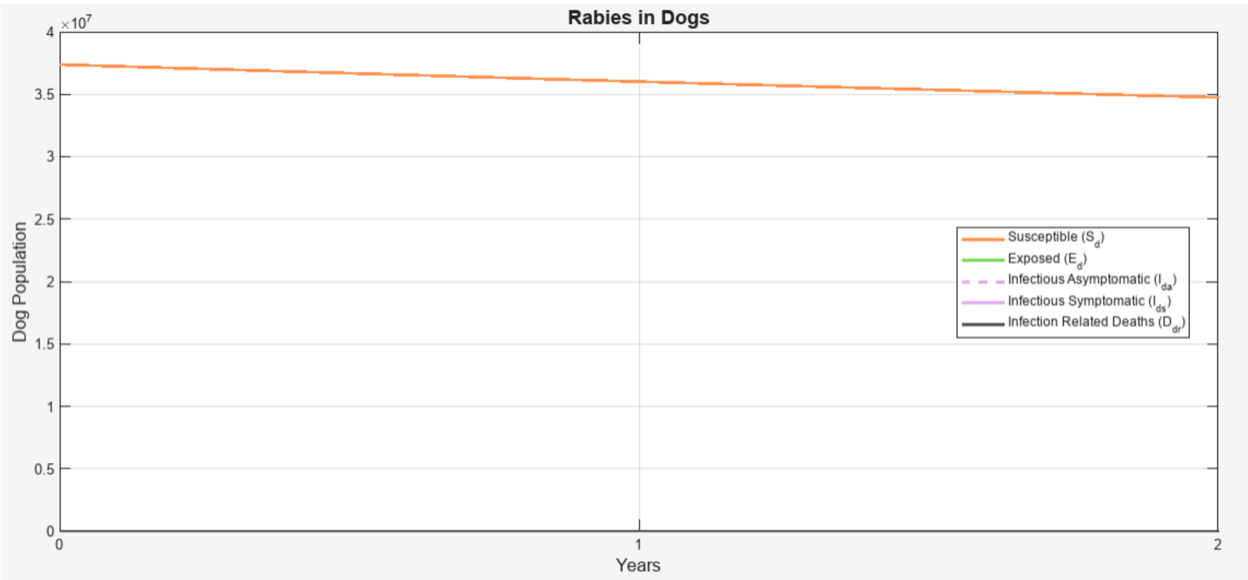

Figure 9: SIR for Dogs

We also ran this simulation while excluding the susceptible populations from our plot. Figures y & y+1 show us how the infection spreads and how many deaths result from rabies in each population respectively. Over 2 years we are seeing ~3,000 dog deaths and ~59,000 human deaths. As discussed above, we fit our parameters assuming ~25,222 deaths/year in India. Comparing this number with our model output confirms a good fit.

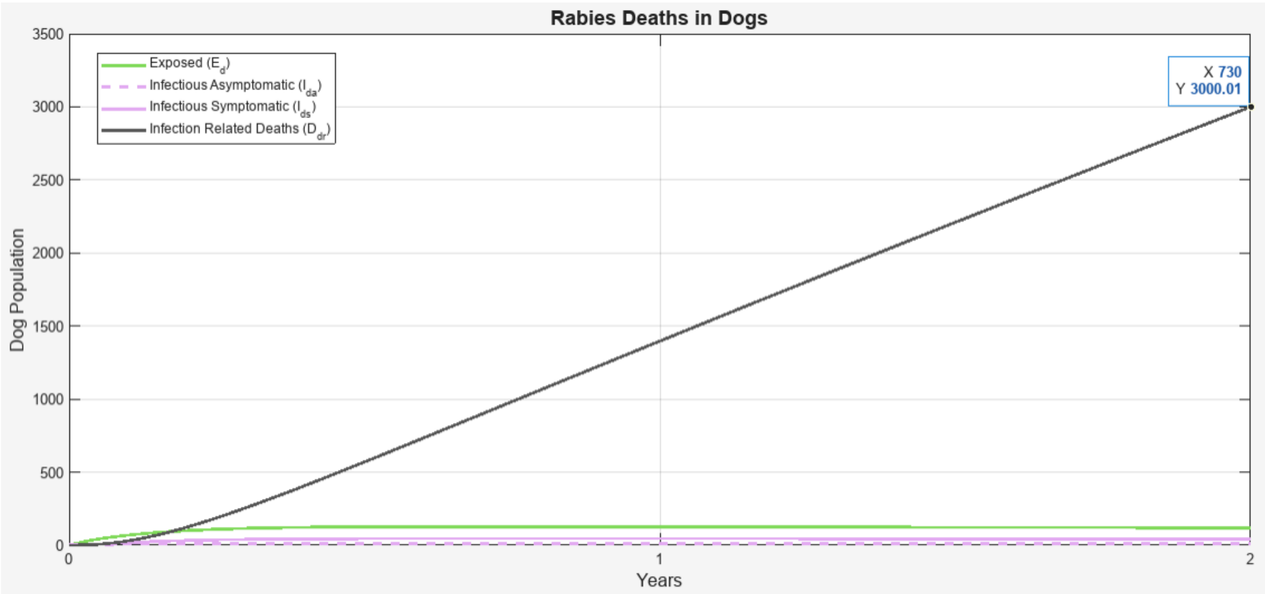

Figure 10: Rabies Death in Dogs

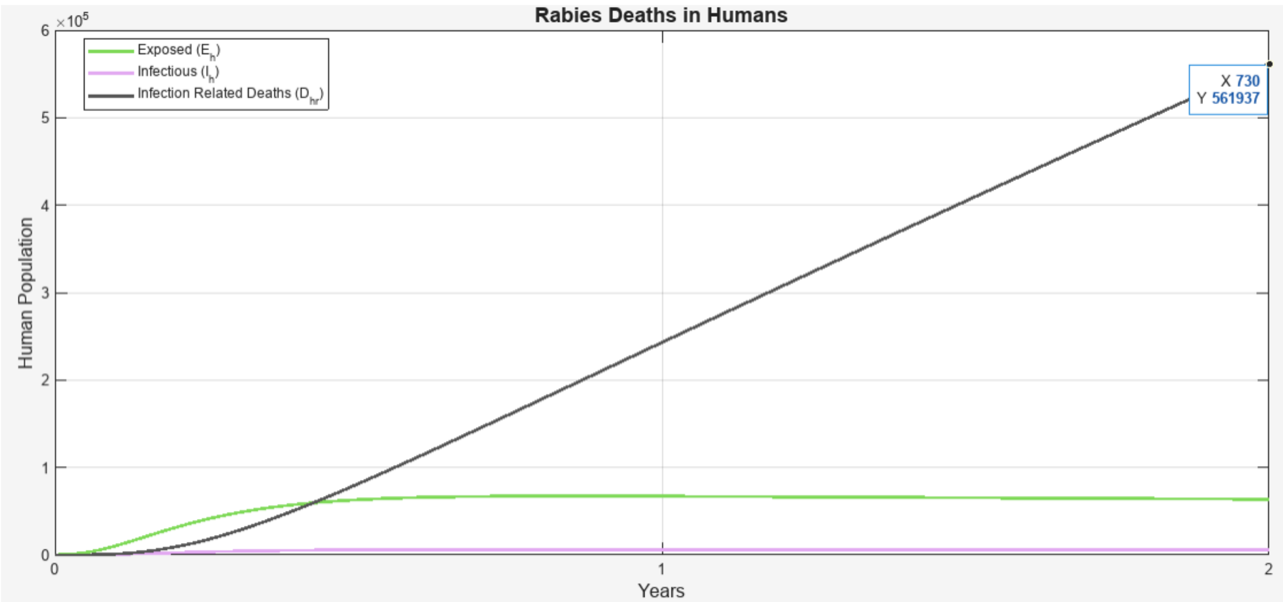

Figure 11: Rabies Death in Humans

4.3: Sensitivity Analysis

Sensitivity analysis over a 20 year period helps provide insight into the most impactful parameters in our model and how they progress over time. Since we chose to treat most of our parameters as constant they appear as straight lines, however, we see one parameter, namely, the birth rate of stray dogs, appears to increase as time goes on. Indicating its impact on the number of human lives is dependent on time. We also see that as the symptomatic transmission rate from dogs to humans increases the number of human deaths increases, and the opposite effect for the treatment rate of humans. Finally we can conclude that the asymptomatic transmission rate and the susceptible human vaccination rate do not make a major impact in the final outcome of our model.

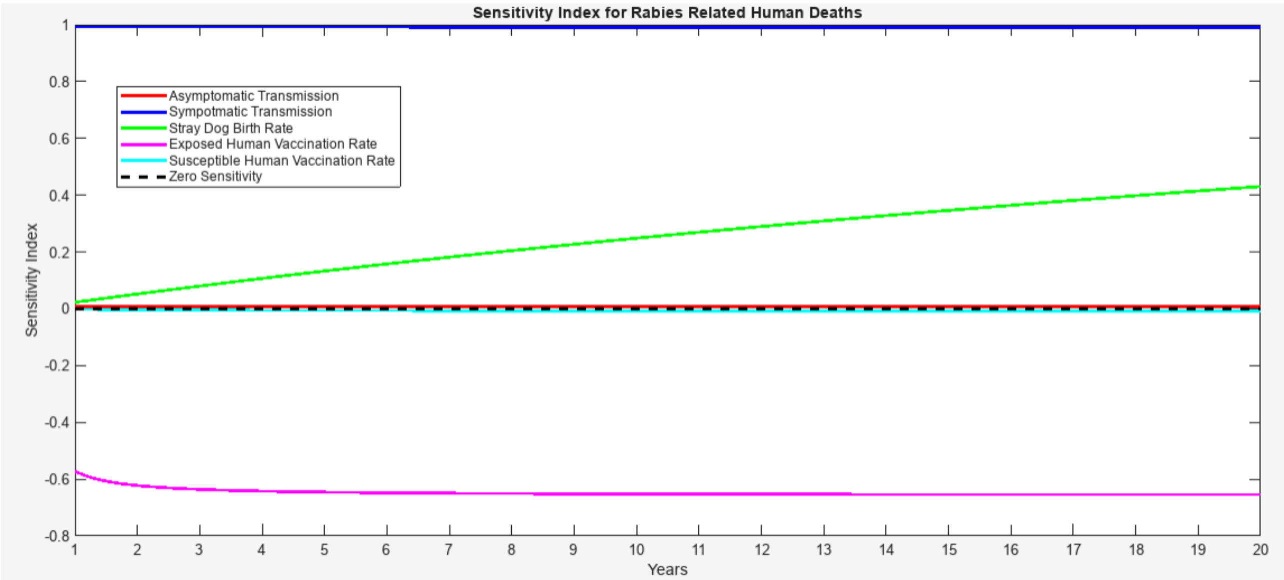

Figure 12: Sensitivity Analysis of Rabies Death for Humans

5. Discussion

5.1: Parameter Fitting

The Bayesian parameter fitting approach employed in this study provided effective estimates for rabies transmission rates, which are particularly difficult to measure in real world settings. The fitting process targeted the estimated annual rabies mortality in India as the primary calibration point. This methodology allowed us to generate plausible parameter values consistent with observed mortality patterns consistent with global reports. A major hurdle in parameter estimation was that the model was both structurally and practically non-identifiable.

The posterior distributions for the fitted parameters displayed multiple local maxima (“spikes”), demonstrating that several combinations of transmission rates can generate similar model outputs. This reflects a limitation common in infectious disease models with scarce data. India lacks robust surveillance systems for rabies with human cases often underreported, there is insufficient data to uniquely constrain all parameters. Therefore, our estimated transmission rates should be interpreted as representative values consistent with observed mortality patterns. The simplifications required for calibration raises another limitation as only a single key data point (annual deaths by rabies) was used for fitting. The model cannot capture more complex patterns such as seasonal variation, regional heterogeneity, or fluctuating vaccination coverage. A more data rich approach using dog bite incidence, confirmed cases in animals, and time series human case data would improve identifiability and likely reduce uncertainty.

Despite these limitations, the fitted parameters produced outputs closely aligned with the expected mortality burden in India, validating its use for scenario analysis and supporting the internal consistency of the model.

5.2: Model Results

The model outputs for both dog and human populations show several important epidemiological insights.

Rabies does not substantially affect the long term population dynamics of humans or stray dogs, even when simulated over multiple years. The disease is slow to spread due to the combination of a low number of infected dogs, a short infectious window, and an inherently slow transmission rate among the dog population. Rabies continues to have a low incidence but high death rate, consistent with observations from endemic regions.

The model forecasts a substantial human death toll of around 56,000 within a two-year span, even with a low prevalence of infected dogs at any single moment.

The model forecasts a substantial human death toll of around 56,000 within a two-year span, even with a low prevalence of infected dogs at any single moment. This aligns with the widely reported global estimate that India accounts for 36% of global human rabies deaths annually [11]. The discrepancy between low dog infection prevalence and high human mortality highlights the disproportionate risk posed by symptomatic dogs and the high frequency of human–dog interactions in densely populated urban environments.

Comparing infected and symptomatic dogs underscores the significance of clinical progression; the minimal time dogs spend in the asymptomatic infectious stage suggests asymptomatic transmission has a limited impact, which aligns with our sensitivity analysis. Given that the majority of human cases are caused by symptomatic dogs, it is vital to boost public awareness and ensure a swift, organized response following the observation of a rabid dog in a community.

The model outputs emphasize the effectiveness of human Post-Exposure Prophylaxis (PEP). In scenarios where treatment rates increase, human deaths decrease sharply, reflecting real-world evidence that prompt PEP is nearly 100% effective [12]. This indicates that improving access to PEP, especially in rural regions, remains one of the most actionable strategies for reducing mortality.

5.3: Sensitivity Analysis

The sensitivity analysis highlights the parameters that play the most significant role in shaping human mortality outcomes. Several clear trends emerged:

1. The stray dog birth rate has the strongest long-term effect on population size. Because its influence grows steadily over time, effective dog population management, mainly through sterilization, can significantly reduce the future rabies burden. This strategy is supported by positive outcomes observed in Indian cities, where ongoing sterilization efforts have successfully decreased both the number of stray dogs and the incidence of dog bites.[13]
2. Symptomatic dog-to-human transmission (β_4) is a major driver of human mortality, with predicted deaths rising proportionally to this rate. Since symptomatic dogs are prone to aggressive behavior and biting, community education and the rapid containment of suspected rabid dogs are essential steps to significantly reduce human exposure.
3. The human Post-Exposure Prophylaxis (PEP) treatment rate (ψ_3) represents one of the most effective intervention points. Even small increases in the proportion of exposed individuals who receive timely treatment led to dramatic reductions in mortality, reinforcing the WHO recommendation that universal access to PEP is a pillar of rabies control.[14]
4. Asymptomatic transmission (β_1) has minimal influence on overall dynamics due to the very short duration of the infectious stage in dogs (approximately 10 days) [15]. This finding supports the current public health strategy of focusing primarily on symptomatic animals.
5. Vaccination of susceptible humans (ψ_4) also has limited effect because rabies vaccination for humans requires multiple doses and offers protection for only approximately 2 years, large-scale preventive vaccination campaigns are not cost-effective in endemic regions. Instead, resources are better allocated toward PEP availability, dog vaccination (in real-world programs), and dog population control [16].

The sensitivity analysis confirms that the most effective strategies for reducing human mortality are targeting symptomatic dog behavior, ensuring rapid Post-Exposure Prophylaxis (PEP) administration, and achieving long-term reductions in the stray dog population.

6. Conclusion

Rabies is an extremely deadly disease that has been impacting humans for thousands of years. The majority of infections are transmitted to humans from stray dogs, making the disease much more deadly in developing countries, where there are many stray dogs and less protective measures in place for people. This modelling exercise aims to uncover the main factors that drive rabies transmission and what measures will be most effective in protecting the public from such a dangerous disease. Our model showed that rabies is not the type of disease that can take over a population (figures 9 & 10), however, it is still projected to kill 56,000 people in India alone over 2 years (figure 12). The results from our sensitivity analysis showed us that the most effective measures that can be taken to reduce deaths are:

1. Efforts to sterilize stray dogs have a long-term impact. As time goes on, the population of stray dogs will slowly decline, reducing the number of vectors that can transmit rabies to humans.
2. Efforts to increase public awareness of rabies, which result in reducing the transmission rate from symptomatic dogs to humans, has the strongest potential to reduce the number of human lives lost.
3. Increasing the rate at which exposed humans seek PEP treatment after being bitten also plays a major role in reducing the number of deaths. Since the treatment methods are highly effective, as more people seek treatments, the number of deaths will decrease rapidly.
4. Asymptomatic transmission rates and the rate at which susceptible humans are vaccinated do not have a relatively strong impact on the overall number of deaths. Both of these effects are small due to
    a. The low amount of time infected dogs spend asymptomatic (~1 day) limits the number of transmissive events they will be involved in.
    b. While the rabies vaccine is nearly 100% effective, it also only lasts for ~2 years on average, meaning widespread vaccination efforts would not be worth the effort.
5. Due to limited data of transmission, we needed to do Bayesian parameter fitting on all 5 β values (modes of transmission). However, the predicted values gave expected results as all other parameters were identified as the real life parameters.

Our model accurately predicts the number of human deaths expected in India over a 2 year period and provides clear direction on next steps a government or organization can take to begin reducing the impact of rabies on human lives.

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8. Appendix

MATLAB software was used for coding purposes. Following are the codes (2 scripts and 2 functions).

8.1: Rabies Main Script

```
% MATH 574: Mathematical Modelling of Infectious Diseases
% Project Title: The Impact of Rabies in Developing Countries
% Project Group # 1: Sharma Group
% Date: November 23, 2025
%% ---------------------------------------------------------
%  Run Rabies Model + Produce Human & Dog Plots (with Deaths)
%  ---------------------------------------------------------
clear; clc;
%% 0. Simulation time
years = 20; % years
time = 365*years; % days
t = linspace(0,time,2981);
%% 1. Initial conditions (must match 10 compartments)
% Order:
% S_d, E_d, I_da, I_ds, D_dr, S_h, V_h, E_h, I_h, D_hr
x0 = [
   37379597    % S_d
   0           % E_d
   1           % I_da
   2           % I_ds
   0           % D_dr
   1.5e9*0.99  % S_h - 99% of the total population does not vaccinate
   1.5e9*0.01  % V_h - 1% of the total population does vaccinate
   0           % E_h
   0           % I_h
   0           % D_hr
];
%% 2. Parameter estimates (daily rates)
%Beta MAP Estimates (calculated through Bayesian Fitting - first run)
%See codes for loglike_rabies_model.m and parameter_fitting_v1.m
%beta1 = 5.5185e-10; asymptomatic dog -> dog transmission
%beta2 = 8.4061e-10; symptomatic dog -> dog transmission
%beta3 = 8.2878e-10; asymptomatic dog -> human transmission
%beta4 = 3.3016e-08; symptomatic dog -> human transmission
%beta5 = 8.0247e-08; background infection (other animals -> dog
transmission)
% Human parameters
delta = 1/(2*365);          % loss of immunity (2 years)
beta3 = 8.2878e-10;         % asymptomatic dog -> human transmission
beta4 = 3.3016e-08;         % symptomatic dog -> human transmission
alpha2 = 1/(70.82*365);     % human natural mortality (avg. lifespan 70.82
years)
tao2 = 1/(2.5*30);          % human incubation rate (2.5 months)
```

```
psi1 = (0.60/0.40)*(tao2); % 60% of the exposed people finish at least 3
dozes of vaccination
psi2 = 0.005/365;          % susceptible vaccination rate (0.5%)
phi2 = 1/7;                % human rabies-induced mortality rate after
symptom onset (mean ~7 days)
N0_h = x0(6) + x0(7) + x0(8) + x0(9) + x0(10); %total human population
A = 0.01655*N0_h/365;      % human recruitment (births)
% Dog parameters
beta1 = 5.5185e-10;        % asymptomatic dog -> dog transmission
beta2 = 8.4061e-10;        % symptomatic dog -> dog transmission
beta5 = 8.0247e-08;        % background infection (other animals -> dog
transmission)
alpha1 = 1/(13*365);       % natural dog mortality (avg. lifespan ~13 years
for many free-roaming dogs)
tao1 = 1/(4*7);            % dog incubation rate (4 weeks)
omega = 1/3;               % asymptomatic -> symptomatic rate (mean 3 days
once moving to clinical phase)
phi1 = 1/10;               % symptomatic dog death (rabies) (mean 10 days
after symptoms)
N0_d = x0(1) + x0(2) + x0(3) + x0(4) + x0(5); %total dog population
B = 0.039*N0_d/365;        % dog recruitment (births)
%% 3. Run the model
par_human = [delta, beta3, beta4, alpha2, tao2, psi1, psi2, phi2, A];
%array for all human parameters
par_dog = [beta1, beta2, beta5, alpha1, tao1, omega, phi1, B]; %array for
all dog parameters
sol = rabies_model_v2(t, par_dog, par_human, x0); %solution from model
function
%% 4. Compartment Color Coding For Plots
S_col = [255 145 77] / 255;   % Susceptible - Orange
V_col = [92 225 230] / 255;   % Vaccinated - Cyan
E_col = [126 217 87] / 255;   % Exposed - Green
I_col = [226 169 241] / 255;  % Infectious - Purple
D_col = [84 84 84] / 255;     % Dead - Grey
%% ----------------
%  5a. Plot Humans
%  ----------------
figure;
plot(sol.T, sol.S_h, 'Color', S_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.V_h, 'Color', V_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.E_h, 'Color', E_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.I_h, 'Color', I_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.D_hr, 'Color', D_col, 'LineWidth', 2); hold off;
xlabel('Years', 'FontSize', 12);
ylabel('Human Population', 'FontSize', 12);
title('Rabies in Humans', 'FontSize', 14);
xticks(0:365:730);
xticklabels(0:2);
xlim([0 730]);
legend({'Susceptible (S_h)', ...
       'Vaccinated (V_h)', ...
       'Exposed (E_h)', ...
       'Infectious (I_h)', ...
       'Infection Related Deaths (D_{hr})'}, ...
       'Location', 'best');
```

```
grid on;
%% --------------
%  6a. Plot Dogs
%  --------------
figure;
plot(sol.T, sol.S_d, 'Color', S_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.E_d, 'Color', E_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.I_da, '--', 'Color', I_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.I_ds, 'Color', I_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.D_dr,  'Color', D_col, 'LineWidth', 2); hold off;
xlabel('Years', 'FontSize', 12);
ylabel('Dog Population', 'FontSize', 12);
title('Rabies in Dogs', 'FontSize', 14);
xticks(0:365:730);
xticklabels(0:2);
xlim([0 730]);
legend({'Susceptible (S_d)', ...
       'Exposed (E_d)', ...
       'Infectious Asymptomatic (I_{da})', ...
       'Infectious Symptomatic (I_{ds})', ...
       'Infection Related Deaths (D_{dr})'}, ...
       'Location', 'best');
grid on;
%% ----------------------
%  5b. Plot Human Deaths
%  ----------------------
figure;
plot(sol.T, sol.E_h, 'Color', E_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.I_h, 'Color', I_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.D_hr, 'Color', D_col, 'LineWidth', 2); hold off
xlabel('Years', 'FontSize', 12);
ylabel('Human Population', 'FontSize', 12);
title('Rabies Deaths in Humans', 'FontSize', 14);
xticks(0:365:730);
xticklabels(0:2);
xlim([0 730]);
legend({'Exposed (E_h)', ...
       'Infectious (I_h)', ...
       'Infection Related Deaths (D_{hr})'}, ...
       'Location', 'best');
grid on;
%% --------------------
%  6b. Plot Dog Deaths
%  --------------------
figure;
plot(sol.T, sol.E_d, 'Color', E_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.I_da, '--', 'Color', I_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.I_ds, 'Color', I_col, 'LineWidth', 2); hold on;
plot(sol.T, sol.D_dr, 'Color', D_col, 'LineWidth', 2); hold off;
xlabel('Years', 'FontSize', 12);
ylabel('Dog Population', 'FontSize', 12);
title('Rabies Deaths in Dogs', 'FontSize', 14);
xticks(0:365:730);
xticklabels(0:2);
xlim([0 730]);
```

```
legend({'Exposed (E_d)', ...
        'Infectious Asymptomatic (I_{da})', ...
        'Infectious Symptomatic (I_{ds})', ...
        'Infection Related Deaths (D_{dr})'}, ...
        'Location', 'best');
grid on;
%% ------------------------
%  7. Sensitivity Analysis
%  ------------------------
%Assume the following changes are being investigated for parameter
sensitivity
p = 0.05;  % 5% for dog -> human transmission rates variability (beta3,
beta4)
p1 = 0.25; % 25% for dog birth rate variability (B)
p2 = 0.5;  % 50% for exposed human vaccination rate variability (psi1)
p3 = 0.10; % 10% for susceptible human vaccination rate variability (psi2)
%Sensitivity index for "beta3"
beta3_low = (1-p)*beta3;
beta3_high = (1+p)*beta3;
%Calculate model outcomes for "beta3" sensitivity
par_human_low = [delta, beta3_low, beta4, alpha2, tao2, psi1, psi2, phi2,
A]; %array for all human parameters
[sol_low] = rabies_model_v2(t, par_dog, par_human_low, x0); %solution from
model function
par_human_high = [delta, beta3_high, beta4, alpha2, tao2, psi1, psi2, phi2,
A]; %array for all human parameters
[sol_high] = rabies_model_v2(t, par_dog, par_human_high, x0); %solution
from model function
%for Model Outcome D_hr (Human Deaths - Rabies Related)
gamma_beta3 = ((sol_high.D_hr - sol_low.D_hr)./(2*beta3*p)) .*
(beta3./sol.D_hr);
%Sensitivity index for "beta4"
beta4_low = (1-p)*beta4;
beta4_high = (1+p)*beta4;
%Calculate model outcomes for "beta4" sensitivity
par_human_low = [delta, beta3, beta4_low, alpha2, tao2, psi1, psi2, phi2,
A]; %array for all human parameters
[sol_low] = rabies_model_v2(t, par_dog, par_human_low, x0); %solution from
model function
par_human_high = [delta, beta3, beta4_high, alpha2, tao2, psi1, psi2, phi2,
A]; %array for all human parameters
[sol_high] = rabies_model_v2(t, par_dog, par_human_high, x0); %solution
from model function
%for Model Outcome D_hr (Human Deaths - Rabies Related)
gamma_beta4 = ((sol_high.D_hr - sol_low.D_hr)./(2*beta4*p)) .*
(beta4./sol.D_hr);
%Sensitivity index for "B"
B_low = (1-p1)*B;
B_high = (1+p1)*B;
%Calculate model outcomes for "B" sensitivity
par_dog_low = [beta1, beta2, beta5, alpha1, tao1, omega, phi1, B_low];
%array for all dog parameters
[sol_low] = rabies_model_v2(t, par_dog_low, par_human, x0); %solution from
model function
```

```
par_dog_high = [beta1, beta2, beta5, alpha1, tao1, omega, phi1, B_high];
%array for all dog parameters
[sol_high] = rabies_model_v2(t, par_dog_high, par_human, x0); %solution
from model function
%for Model Outcome D_hr (Human Deaths - Rabies Related)
gamma_B = ((sol_high.D_hr - sol_low.D_hr)./(2*B*p1)) .* (B./sol.D_hr);
%Sensitivity index for "psi1"
psi1_low = (1-p2)*psi1;
psi1_high = (1+p2)*psi1;
%Calculate model outcomes for "psi1" sensitivity
par_human_low = [delta, beta3, beta4, alpha2, tao2, psi1_low, psi2, phi2,
A]; %array for all human parameters
[sol_low] = rabies_model_v2(t, par_dog, par_human_low, x0); %solution from
model function
par_human_high = [delta, beta3, beta4, alpha2, tao2, psi1_high, psi2, phi2,
A]; %array for all human parameters
[sol_high] = rabies_model_v2(t, par_dog, par_human_high, x0); %solution
from model function
%for Model Outcome D_hr (Human Deaths - Rabies Related)
gamma_psi1 = ((sol_high.D_hr - sol_low.D_hr)./(2*psi1*p2)) .*
(psi1./sol.D_hr);
%Sensitivity index for "psi2"
psi2_low = (1-p3)*psi2;
psi2_high = (1+p3)*psi2;
%Calculate model outcomes for "psi2" sensitivity
par_human_low = [delta, beta3, beta4, alpha2, tao2, psi1, psi2_low, phi2,
A]; %array for all human parameters
[sol_low] = rabies_model_v2(t, par_dog, par_human_low, x0); %solution from
model function
par_human_high = [delta, beta3, beta4, alpha2, tao2, psi1, psi2_high, phi2,
A]; %array for all human parameters
[sol_high] = rabies_model_v2(t, par_dog, par_human_high, x0); %solution
from model function
%for Model Outcome D_hr (Human Deaths - Rabies Related)
gamma_psi2 = ((sol_high.D_hr - sol_low.D_hr)./(2*psi2*p3)) .*
(psi2./sol.D_hr);
%% ----------------------
%  7b. Sensitivity Plots
%  ----------------------
%for Model Outcome D_hr (Human Deaths - Rabies Related)
figure;
plot(t, transpose(gamma_beta3), '-r', 'LineWidth', 2); hold on;
plot(t, transpose(gamma_beta4), '-b', 'LineWidth', 2); hold on;
plot(t, transpose(gamma_B), '-g', 'LineWidth', 2); hold on;
plot(t, transpose(gamma_psi1), '-m', 'LineWidth', 2); hold on;
plot(t, transpose(gamma_psi2), '-c', 'LineWidth', 2); hold on;
plot(t,zeros(size(t)),'--k', 'LineWidth', 2); hold off;
xlabel('Years');
ylabel('Sensitivity Index');
title('Sensitivity Index for Rabies Related Human Deaths');
xticks(0:365:365*20);
xticklabels(0:20);
xlim([365 365*20]);
```

```
X1 = ({'Asymptomatic Transmission','Sympotmatic Transmission','Stray Dog
Birth Rate','Exposed Human Vaccination Rate','Susceptible Human Vaccination
Rate','Zero Sensitivity'});
legend(X1, ...
       'Location', 'best');
savefig("local_sensitivity_D_hr.fig");
```

8.2: Rabies Model Function

```
% MATH 574: Mathematical Modelling of Infectious Diseases
% Project Title: The Impact of Rabies in Developing Countries
% Project Group # 1: Sharma Group
% Date: November 23, 2025
function sol = rabies_model_v2(t, par_dog, par_human, x0)
% Human parameters
delta = par_human(1);  % loss of immunity (2 years)
beta3 = par_human(2);  % asymptomatic dog -> human transmission
beta4 = par_human(3);  % symptomatic dog -> human transmission
alpha2 = par_human(4); % human natural mortality (avg. lifespan 70.82
years)
tao2 = par_human(5);   % human incubation rate (2.5 months)
psi1 = par_human(6);   % 60% of the exposed people finish at least 3 dozes
of vaccination
psi2 = par_human(7);   % susceptible vaccination rate (0.5%)
phi2 = par_human(8);   % human rabies-induced mortality rate after symptom
onset (mean ~7 days)
A = par_human(9);      % human recruitment (births)
% Dog parameters
beta1 = par_dog(1);    % asymptomatic dog -> dog transmission
beta2 = par_dog(2);    % symptomatic dog -> dog transmission
beta5 = par_dog(3);    % background infection (other animals -> dog
transmission)
alpha1 = par_dog(4);   % natural dog mortality (avg. lifespan ~13 years for
many free-roaming dogs)
tao1 = par_dog(5);     % dog incubation rate (4 weeks)
omega = par_dog(6);    % asymptomatic -> symptomatic rate (mean 3 days once
moving to clinical phase)
phi1 = par_dog(7);     % symptomatic dog death (rabies) (mean 10 days after
symptoms)
B = par_dog(8);        % dog recruitment (births)
%% -----------
%  ODE System
%  -----------
% Dog Compartments
% x(1) = S_d
% x(2) = E_d
% x(3) = I_da
% x(4) = I_ds
% x(5) = D_dr
% Human Compartments
```

```
% x(6) = S_h
% x(7) = V_h
% x(8) = E_h
% x(9) = I_h
% x(10) = D_hr
H = @(t,x) [
   % ----- Dogs -----
    B - beta1*x(1)*x(3) - beta2*x(1)*x(4) - beta5*x(1) - alpha1*x(1);
    beta1*x(1)*x(3) + beta2*x(1)*x(4) + beta5*x(1) - (tao1+alpha1)*x(2);
    tao1*x(2) - (omega+alpha1)*x(3);
    omega*x(3) - (phi1+alpha1)*x(4);
    phi1*x(4);
    % ----- Humans -----
    A + delta*x(7) - beta3*x(6)*x(3) - beta4*x(6)*x(4) -
(psi2+alpha2)*x(6);
    psi1*x(8) + psi2*x(6) - (delta+alpha2)*x(7);
    beta3*x(6)*x(3) + beta4*x(6)*x(4) - (psi1+tao2+alpha2)*x(8);
    tao2*x(8) - (phi2+alpha2)*x(9);
    phi2*x(9);
];
%% -----------------
%  Solve ODE System
%  -----------------
[T,Y] = ode45(H, t, x0);
%% -----------------
%  Output Structure
%  -----------------
sol.T    = T;
% ----- Dogs -----
sol.S_d  = Y(:,1);
sol.E_d  = Y(:,2);
sol.I_da = Y(:,3);
sol.I_ds = Y(:,4);
sol.D_dr = Y(:,5);
% ----- Humans -----
sol.S_h  = Y(:,6);
sol.V_h  = Y(:,7);
sol.E_h  = Y(:,8);
sol.I_h  = Y(:,9);
sol.D_hr = Y(:,10);
end
```

8.3: PARAMETER FITTING SCRIPT

```
% MATH 574: Mathematical Modelling of Infectious Diseases
% Project Title: The Impact of Rabies in Developing Countries
% Project Group # 1: Sharma Group
% Date: December 03, 2025
%% Parameters to Fit
t = linspace(0,365*20,2981);  % Reduced from 2981 for faster computation
%beta1, beta2, beta3, beta4, and beta5 (transmission rates)
% Define prior ranges for log10 of each beta parameter
widths = [log10(1e-10) log10(1e-7);
         log10(1e-10) log10(1e-7);
         log10(1e-10) log10(1e-7);
         log10(1e-10) log10(1e-7);
         log10(1e-10) log10(1e-7)];
% Expected final death count from data/observations
final_deaths_observed = 23000/0.4;
% Set default plot properties
set(0,'defaultLineLineWidth',1.5);
set(0,'defaultLineMarkerSize',9);
set(0, 'DefaultAxesFontName', 'Arial');
set(0, 'DefaultTextFontName', 'Arial');
% Prior distributions for log10_beta1 through log10_beta5 (uniform)
prior1 = @(log10_beta1) unifpdf(log10_beta1, widths(1,1), widths(1,2));
prior2 = @(log10_beta2) unifpdf(log10_beta2, widths(2,1), widths(2,2));
prior3 = @(log10_beta3) unifpdf(log10_beta3, widths(3,1), widths(3,2));
prior4 = @(log10_beta4) unifpdf(log10_beta4, widths(4,1), widths(4,2));
prior5 = @(log10_beta5) unifpdf(log10_beta5, widths(5,1), widths(5,2));
% Parameter names for tracking
par_names = ["log10(beta1)", "log10(beta2)", "log10(beta3)",
"log10(beta4)", "log10(beta5)"];
%% Parameter Fit Setup
% Given the vector of parameters theta = [log10_beta1, log10_beta2,
log10_beta3, log10_beta4, log10_beta5],
% logl returns the log likelihood distribution
logl = @(theta) loglike_rabies_model(theta, final_deaths_observed);
% 'logp' is the natural logarithm of the prior distribution for the
% parameters. The prior distribution is a product of uniform
% distributions, the resulting joint distribution does integrate to one.
logp = @(theta) log(prior1(theta(1))) + log(prior2(theta(2))) +
log(prior3(theta(3))) + log(prior4(theta(4))) + log(prior5(theta(5)));
theta0 = zeros(5, 5); % Reduced from 10 to 5 chains for faster computation
% Generate the vector of initial conditions
k = 0;
j = 1;
fprintf('Generating initial parameter values for chains...\n');
while(j < (5+1))  % Changed from (10+1) to (5+1)
   while(k == 0 || isnan(k) || isinf(k))
       initial = [widths(1,1)+(widths(1,2)-widths(1,1))*rand ...
                  widths(2,1)+(widths(2,2)-widths(2,1))*rand ...
                  widths(3,1)+(widths(3,2)-widths(3,1))*rand ...
                  widths(4,1)+(widths(4,2)-widths(4,1))*rand ...
                  widths(5,1)+(widths(5,2)-widths(5,1))*rand];
```

```
        k = logl(initial) + logp(initial);
    end
    theta0(1,j) = initial(1);
    theta0(2,j) = initial(2);
    theta0(3,j) = initial(3);
    theta0(4,j) = initial(4);
    theta0(5,j) = initial(5);
    j = j + 1;
    k = 0;
    fprintf('Chain %d initialized\n', j-1);
end
% 'logfuns' holds one function for the log of the priors and another
% function for the log of the likelihood function
logfuns = {@(theta)logp(theta) @(theta)logl(theta)};
%% Affine Fitting Algorithm
% 'gwmcmc' runs the MCMC sampler with affine invariance
fprintf('Starting MCMC sampling...\n');
[models, logP] = gwmcmc(theta0, logfuns, 50000);  % Reduced from 1000000
for faster computation
% 'models' holds the proposed guesses for the parameter values
save Project_models.mat models
fprintf('Models saved to Project_models.mat\n');
% 'logP' holds the corresponding value of the log prior and the value of
the
% log likelihood for each guess of the parameter values
save Project_logP.mat logP
fprintf('Log probabilities saved to Project_logP.mat\n');
fprintf('MCMC sampling complete!\n');
%% Assess results
% Let 'l' hold the values of the log prior and log likelihood
l = logP;
% Check the size and determine appropriate burn-in
[~, n_samples, n_chains] = size(l);
fprintf('\nMCMC Results: %d samples x %d chains\n', n_samples, n_chains);
% Set burn-in (10% of samples, or 5000, whichever is smaller)
burnin = min(5000, floor(0.1 * n_samples));
fprintf('Using burn-in of %d samples per chain\n', burnin);
% Remove burn-in samples
if burnin > 0 && burnin < n_samples
    l(:,:,1:burnin)=[];
    fprintf('Burn-in removed successfully\n');
else
    warning('Burn-in size inappropriate - keeping all samples');
end
% Extract the values of the log prior
lprior = l(1,:,:);
% Collapse the samples of all chains into a single larger sample
lprior = lprior(:,:);
% Extract the values of the log likelihood
llik = l(2,:,:);
% Collapse the samples of all chains into a single larger sample
llik = llik(:,:);
% Add the log prior and log likelihood together to get the log unnormalized
% posterior samples
l_un_post_samples = lprior + llik;
```

```
% Let 'm' hold the values of the proposed guesses for the parameter values
m = models;
% Check the size and remove burn-in from models
[~, n_samples_m, n_chains_m] = size(m);
if burnin > 0 && burnin < n_samples_m
   m(:,:,1:burnin)=[];
   fprintf('Burn-in removed from parameter samples\n');
else
   warning('Burn-in size inappropriate for models - keeping all samples');
end
% Collapse the samples of all chains into a single larger sample
m = m(:,:);
theta_samples = m;
% Display final sample size
fprintf('Total posterior samples after burn-in: %d\n', size(theta_samples,
2));
% Extract proposed values for each parameter
% 'k1' contains all the proposed values for parameter 'log10_beta1'
k1 = m(1,:);
% 'k2' contains all the proposed values for parameter 'log10_beta2'
k2 = m(2,:);
% 'k3' contains all the proposed values for parameter 'log10_beta3'
k3 = m(3,:);
% 'k4' contains all the proposed values for parameter 'log10_beta4'
k4 = m(4,:);
% 'k5' contains all the proposed values for parameter 'log10_beta5'
k5 = m(5,:);
% Parameter values at the maximum log unnormalized posterior value
[M, I] = max(l_un_post_samples);
%% Parameter Posterior Plots
% log10(beta1)
figure;
post_silhouette_plot(theta_samples, l_un_post_samples, par_names, 1)
savefig("post_surf_log10(beta1).fig");
[l_HPD, u_HPD] = HPD(10.^k1, 0.05);
k1_atmax = 10.^theta_samples(1,I);
k1_median = 10.^median(k1);
fprintf('Beta1 - 95%% CI: (%.4e, %.4e)\n', l_HPD, u_HPD);
fprintf('Beta1 - MAP estimate: %.4e\n', k1_atmax);
fprintf('Beta1 - Median: %.4e\n\n', k1_median);
% log10(beta2)
figure;
post_silhouette_plot(theta_samples, l_un_post_samples, par_names, 2)
savefig("post_surf_log10(beta2).fig");
[l_HPD, u_HPD] = HPD(10.^k2, 0.05);
k2_atmax = 10.^theta_samples(2,I);
k2_median = 10.^median(k2);
fprintf('Beta2 - 95%% CI: (%.4e, %.4e)\n', l_HPD, u_HPD);
fprintf('Beta2 - MAP estimate: %.4e\n', k2_atmax);
fprintf('Beta2 - Median: %.4e\n\n', k2_median);
% log10(beta3)
figure;
post_silhouette_plot(theta_samples, l_un_post_samples, par_names, 3)
savefig("post_surf_log10(beta3).fig");
[l_HPD, u_HPD] = HPD(10.^k3, 0.05);
```

```
k3_atmax = 10.^theta_samples(3,I);
k3_median = 10.^median(k3);
fprintf('Beta3 - 95%% CI: (%.4e, %.4e)\n', l_HPD, u_HPD);
fprintf('Beta3 - MAP estimate: %.4e\n', k3_atmax);
fprintf('Beta3 - Median: %.4e\n\n', k3_median);
% log10(beta4)
figure;
post_silhouette_plot(theta_samples, l_un_post_samples, par_names, 4)
savefig("post_surf_log10(beta4).fig");
[l_HPD, u_HPD] = HPD(10.^k4, 0.05);
k4_atmax = 10.^theta_samples(4,I);
k4_median = 10.^median(k4);
fprintf('Beta4 - 95%% CI: (%.4e, %.4e)\n', l_HPD, u_HPD);
fprintf('Beta4 - MAP estimate: %.4e\n', k4_atmax);
fprintf('Beta4 - Median: %.4e\n\n', k4_median);
% log10(beta5)
figure;
post_silhouette_plot(theta_samples, l_un_post_samples, par_names, 5)
savefig("post_surf_log10(beta5).fig");
[l_HPD, u_HPD] = HPD(10.^k5, 0.05);
k5_atmax = 10.^theta_samples(5,I);
k5_median = 10.^median(k5);
fprintf('Beta5 - 95%% CI: (%.4e, %.4e)\n', l_HPD, u_HPD);
fprintf('Beta5 - MAP estimate: %.4e\n', k5_atmax);
fprintf('Beta5 - Median: %.4e\n\n', k5_median);%% Plot Maximum A Posteriori
(MAP) Solution
%Initial conditions: S_d, E_d, I_da, I_ds, D_dr, S_h, V_h, E_h, I_h, D_hr
x0 = [
   37379597    % S_d
   0           % E_d
   1           % I_da
   2           % I_ds
   0           % D_dr
   1.5e9*0.99  % S_h - 99% of the total population does not vaccinate
   1.5e9*0.01  % V_h - 1% of the total population does vaccinate
   0           % E_h
   0           % I_h
   0           % D_hr
];
% Fixed parameters
delta = 1/(2*365);          % loss of immunity (2 years)
alpha2 = 1/(70.82*365);     % human natural mortality (avg. lifespan 70.82
years)
tao2 = 1/(2.5*30);          % human incubation rate (2.5 months)
psi1 = (0.60/0.40)*(tao2); % 60% of the exposed people finish at least 3
dozes of vaccination
psi2 = 0.005/365;           % susceptible vaccination rate (0.5%)
phi2 = 1/7;                 % human rabies-induced mortality rate after
symptom onset (mean ~7 days)
N0_h = x0(6) + x0(7) + x0(8) + x0(9) + x0(10); %total human population
A = 0.01655*N0_h/365;       % human recruitment (births)
% Dog parameters
alpha1 = 1/(13*365);        % natural dog mortality (avg. lifespan ~13 years
for many free-roaming dogs)
tao1 = 1/(4*7);             % dog incubation rate (4 weeks)
```

```
omega = 1/3;                % asymptomatic -> symptomatic rate (mean 3 days
once moving to clinical phase)
phi1 = 1/10;                % symptomatic dog death (rabies) (mean 10 days
after symptoms)
N0_d = x0(1) + x0(2) + x0(3) + x0(4) + x0(5); %total dog population
B = 0.039*N0_d/365;         % dog recruitment (births)
% Extract MAP parameter estimates
beta1_MAP = 10^theta_samples(1,I);
beta2_MAP = 10^theta_samples(2,I);
beta3_MAP = 10^theta_samples(3,I);
beta4_MAP = 10^theta_samples(4,I);
beta5_MAP = 10^theta_samples(5,I);
% Set up parameter arrays with MAP estimates
par_human_MAP = [delta, beta3_MAP, beta4_MAP, alpha2, tao2, psi1, psi2,
phi2, A]; %array for all human parameters
par_dog_MAP = [beta1_MAP, beta2_MAP, beta5_MAP, alpha1, tao1, omega, phi1,
B]; %array for all dog parameters
% Run model with MAP parameters
sol_MAP = rabies_model_v2(t, par_dog_MAP, par_human_MAP, x0);
% Plot results
figure;
hold on
set(gca, 'FontSize', 10, 'LineWidth', 1);
% Plot observed data point at final time
plot(t(365), final_deaths_observed, 'ko', 'MarkerSize', 10, 'DisplayName',
'Observed Deaths')
% Plot model prediction for cumulative human deaths
plot(sol_MAP.T/365, sol_MAP.D_hr, 'b-', 'LineWidth', 2, 'DisplayName', 'MAP
Model Fit')
xlabel('Years'), ylabel('Cumulative Human Deaths from Rabies')
legend('Location', 'best')
title('Maximum A Posteriori Model Fit')
savefig("max_post_solution.fig");
%close all
%% Posterior Predictive Distribution
% Initialize storage arrays
n_samples = length(l_un_post_samples);
final_deaths_predict = zeros(1, n_samples);
discrepancy = zeros(1, n_samples);
discrepancy_pred = zeros(1, n_samples);
ind_D_pred_exceeds = zeros(1, n_samples);
h = waitbar(0, 'Generating posterior predictive samples...');
for i = 1:n_samples
   % Extract current parameter sample
   theta = theta_samples(:, i);
   beta1_i = 10^theta(1);
   beta2_i = 10^theta(2);
   beta3_i = 10^theta(3);
   beta4_i = 10^theta(4);
   beta5_i = 10^theta(5);

   % Set up parameter arrays
   par_human_i = [delta, beta3_i, beta4_i, alpha2, tao2, psi1, psi2, phi2,
A]; %array for all human parameters
```

```
   par_dog_i = [beta1_i, beta2_i, beta5_i, alpha1, tao1, omega, phi1, B];
%array for all dog parameters
   % Run model with current parameters
   sol_i = rabies_model_v2(t, par_dog_i, par_human_i, x0);

   % Extract final cumulative human deaths from model
   final_deaths_model = sol_i.D_hr(end);

   % Calculate discrepancy between observed and model prediction
   p = 0.9;  % Same dispersion parameter as in likelihood
   variance_model = final_deaths_model * (1/p);
   discrepancy(1,i) = ((final_deaths_observed - final_deaths_model)^2) /
variance_model;

   % Generate posterior predictive sample
   r = (final_deaths_model * p) / (1 - p);
   final_deaths_predict(1,i) = nbinrnd(r, p);

   % Calculate predictive discrepancy
   discrepancy_pred(1,i) = ((final_deaths_predict(1,i) -
final_deaths_model)^2) / variance_model;

   % Check if predictive discrepancy exceeds observed discrepancy
   if discrepancy_pred(1,i) > discrepancy(1,i)
       ind_D_pred_exceeds(1,i) = 1;
   else
       ind_D_pred_exceeds(1,i) = 0;
   end

   % Update waitbar
   if mod(i, 100) == 0
       waitbar(i/n_samples, h, sprintf('%d%%', round((i/n_samples)*100)));
   end
end
close(h)
% Save results
save final_deaths_predict.mat final_deaths_predict
save discrepancy.mat discrepancy
save discrepancy_pred.mat discrepancy_pred
save ind_D_pred_exceeds.mat ind_D_pred_exceeds
% Calculate Bayesian p-value
p_value = sum(ind_D_pred_exceeds) / length(ind_D_pred_exceeds);
fprintf('\n=== Posterior Predictive Check ===\n');
fprintf('Bayesian p-value: %.4f\n', p_value);
fprintf('(p-value close to 0.5 indicates good fit)\n');
fprintf('(p-value < 0.05 or > 0.95 may indicate poor fit)\n');
%% Posterior Predictive Plots
% Compute statistics for final deaths predictions
mean_prediction = mean(final_deaths_predict);
lower_bound = prctile(final_deaths_predict, 2.5);
upper_bound = prctile(final_deaths_predict, 97.5);
% Display prediction summary
fprintf('\n=== Posterior Predictive Summary ===\n');
fprintf('Observed final deaths: %.0f\n', final_deaths_observed);
fprintf('Mean predicted deaths: %.0f\n', mean_prediction);
```

```
fprintf('95%% CI: [%.0f, %.0f]\n', lower_bound, upper_bound);
fprintf('MAP predicted deaths: %.0f\n', sol_MAP.D_hr(end));
% Create comprehensive plot
figure;
hold on;
set(gca, 'FontSize', 10, 'LineWidth', 1);
% Plot MAP model trajectory
plot(sol_MAP.T/365, sol_MAP.D_hr, 'm-', 'LineWidth', 2, 'DisplayName', 'MAP
Model');
% Plot observed data point at final time
plot(t(end)/365, final_deaths_observed, 'ko', 'MarkerSize', 10,
'LineWidth', 2, 'DisplayName', 'Observed Deaths');
% Plot prediction interval for final deaths
final_time_years = t(end)/365;
plot([final_time_years, final_time_years], [lower_bound, upper_bound], 'b-
', 'LineWidth', 3, 'DisplayName', '95% Prediction Interval');
plot(final_time_years, mean_prediction, 'bs', 'MarkerSize', 10,
'MarkerFaceColor', 'b', 'DisplayName', 'Mean Prediction');
xlabel('Years');
ylabel('Cumulative Human Deaths from Rabies');
title('Posterior Predictive Distribution');
legend('Location', 'northwest');
grid on;
savefig("Final_Posterior_Predictive_Plot.fig");
hold off;
% Create histogram of posterior predictions
figure;
histogram(final_deaths_predict, 50, 'Normalization', 'pdf', 'FaceColor',
'b', 'EdgeColor', 'none', 'FaceAlpha', 0.6);
hold on;
xline(final_deaths_observed, 'r--', 'LineWidth', 2, 'DisplayName',
'Observed');
xline(mean_prediction, 'b-', 'LineWidth', 2, 'DisplayName', 'Mean
Prediction');
xlabel('Final Cumulative Deaths');
ylabel('Posterior Predictive Density');
title('Posterior Predictive Distribution of Final Deaths');
legend('Location', 'best');
grid on;
savefig("Posterior_Predictive_Histogram.fig");
hold off;
```

8.4: Log Likelihood Rabies Model

```
% MATH 574: Mathematical Modelling of Infectious Diseases
% Project Title: The Impact of Rabies in Developing Countries
% Project Group # 1: Sharma Group
% Date: December 03, 2025
function [loglike] = loglike_rabies_model(theta, final_deaths_observed)
   % Extract parameters (accounting for log10 transformation)
   time = 365*20;   % days
   t = linspace(0,time,2981);

%Initial conditions: S_d, E_d, I_da, I_ds, D_dr, S_h, V_h, E_h, I_h, D_hr
x0 = [
   37379597    % S_d
   0           % E_d
   1           % I_da
   2           % I_ds
   0           % D_dr
   1.5e9*0.99  % S_h - 99% of the total population does not vaccinate
   1.5e9*0.01  % V_h - 1% of the total population does vaccinate
   0           % E_h
   0           % I_h
   0           % D_hr
];
% Fixed parameters
delta = 1/(2*365);          % loss of immunity (2 years)
alpha2 = 1/(70.82*365);     % human natural mortality (avg. lifespan 70.82
years)
tao2 = 1/(2.5*30);          % human incubation rate (2.5 months)
psi1 = (0.60/0.40)*(tao2); % 60% of the exposed people finish at least 3
dozes of vaccination
psi2 = 0.005/365;           % susceptible vaccination rate (0.5%)
phi2 = 1/7;                 % human rabies-induced mortality rate after
symptom onset (mean ~7 days)
N0_h = x0(6) + x0(7) + x0(8) + x0(9) + x0(10); %total human population
A = 0.01655*N0_h/365;       % human recruitment (births)
% Dog parameters
alpha1 = 1/(13*365);        % natural dog mortality (avg. lifespan ~13 years
for many free-roaming dogs)
tao1 = 1/(4*7);             % dog incubation rate (4 weeks)
omega = 1/3;                % asymptomatic -> symptomatic rate (mean 3 days
once moving to clinical phase)
phi1 = 1/10;                % symptomatic dog death (rabies) (mean 10 days
after symptoms)
N0_d = x0(1) + x0(2) + x0(3) + x0(4) + x0(5); %total dog population
B = 0.039*N0_d/365;         % dog recruitment (births)

% Extract fitted parameters from theta (log10 transformed)
beta1 = 10^theta(1); % asymptomatic dog -> dog transmission
beta2 = 10^theta(2); % symptomatic dog -> dog transmission
beta3 = 10^theta(3); % asymptomatic dog -> human transmission
beta4 = 10^theta(4); % symptomatic dog -> human transmission
```

```
beta5 = 10^theta(5); % background infection (other animals -> dog
transmission)

% Run your rabies compartmental model with these parameters
par_human_fitting = [delta, beta3, beta4, alpha2, tao2, psi1, psi2, phi2,
A]; %array for all human parameters
par_dog_fitting = [beta1, beta2, beta5, alpha1, tao1, omega, phi1, B];
%array for all dog parameters
sol = rabies_model_v2(t, par_dog_fitting, par_human_fitting, x0);

   % Check if model ran successfully
   if isempty(sol) || isempty(sol.D_hr) || any(isnan(sol.D_hr)) ||
any(isinf(sol.D_hr))
       loglike = -Inf;
       return;
   end

   % Extract final cumulative HUMAN deaths from model output
   final_deaths_model = sol.D_hr(365);  % Human deaths compartment

   % Calculate log-likelihood using negative binomial distribution
   p = 0.9; % Dispersion parameter - adjust based on your data

   % Negative binomial log-likelihood
   loglike = log(nbinpdf(final_deaths_observed, (final_deaths_model*p)/(1 -
p), p));

   % Handle edge cases
   if isnan(loglike) || isinf(loglike)
       loglike = -Inf;
   end
end
```